



Elevating theoretical insight and predictive accuracy in business research: Combining PLS-SEM and selected machine learning algorithms

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ABSTRACT

We propose a routine for combining partial least squares-structural equation modeling (PLS-SEM) with selected machine learning (ML) algorithms to exploit the two method's causal-predictive and causal-exploratory capabilities. Triangulating these two methods can improve the predictive accuracy of research models, enhance the understanding of relationships, assist in identifying new relationships and therewith contribute to theorizing. We demonstrate the advantages and challenges of triangulating the two methods on an illustrative example along a four-step-routine: (1) Develop a PLS-SEM on a baseline conceptual model and use its standards to assess measurement model quality and generate latent variables scores. (2) Apply specific ML algorithms on the extracted data to validate relationships and identify new (linear) relationships that may go beyond the initial hypotheses; similarly, assess model advancements in the form of nonlinearities and interaction effects. (3) Evaluate the theoretical plausibility of alternative models. (4) Integrate alternative models in PLS-SEM and compare these with the baseline model using a recently proposed prediction-oriented test procedure in PLS-SEM.

1. Introduction

Partial least squares-structural equation modeling (PLS-SEM) was developed as a causal-predictive research method designed to explain the relationships between theoretical or latent constructs, and to facilitate predictive analysis (Jöreskog & Wold, 1982), which has received growing scholarly attention (e.g., Richter et al., 2016a; Richter et al., 2016b; Sarstedt et al., 2022a). Researchers using PLS-SEM can estimate models based on conceptual ideas and have a rich portfolio of validated metrics, decision-trees and checklists at hand to evaluate their explanatory power and accuracy (e.g., Guenther et al., 2023; Hair et al., 2022; Ringle et al., 2023; Sharma et al., 2023). These routines enable a comprehensive comparison of alternative conceptual models. Such comparison has relevance for many research and decision-making situations that involve alternative or weak theories for understanding a phenomenon, potentially complex associations between input factors predicting or causing an outcome, and/ or different contextual dependencies (e.g., Richter et al., 2022). PLS-SEM requires researchers' active conceptualization and modeling skills to test alternative models that reflect different ideas, complexities, or contexts. This inherently entails a subjective selection process, in which the researcher's

perspective and assumptions influence choices.

We argue that this process can be supported by the causal-discovery capabilities of machine learning (ML) algorithms that employ systematic procedures to identify the most likely model based on statistical criteria and metrics. With the emergence of big data (e.g., e-commerce data in marketing; Hair & Sarstedt, 2021), ML algorithms, such as Artificial Neural Networks (ANN) and Bayesian Networks (BN), have gained popularity for various purposes (e.g., Boto Ferreira et al., 2021; Canhoto, 2021; Garbe & Richter, 2009; Sachan et al., 2020; Sánchez-Franco et al., 2019; Tudoran, 2022). BN notably, facilitate the construction of transparent predictive models by discovering data patterns, aiding in the identification of new relationships, and facilitating the integration of a priori information to enable theory testing. BN can valuably complement PLS-SEM to generate new hypotheses and insights or simply to test the robustness of findings (Bollen & Pearl, 2013; Druzdzel & Simon, 1993; Pearl, 2012a; Pearl, 2012b), which has become a requirement in some guidelines on using PLS-SEM (e.g., Sarstedt et al., 2019). Combining SEM and different ML algorithms has become prominent in fields such as medical or health research, where authors frequently integrate insights from data-driven models into conceptual models (e.g., Kenett et al., 2022; Liew et al., 2023). For example, Liew et al. (2023)

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provide new insight regarding the relationships between poor sleep, psychological factors, and the duration of pain, which influence disabilities. However, in business research, the combination of the two approaches is still rare, driven only by a select group of authors. In the *Journal of Business Research*, which is a leading outlet publishing PLS-SEM studies (e.g., [Cho et al., 2022b](#); [Sarstedt et al., 2022a](#); [Sarstedt et al., 2022b](#)), only two percent of the studies using PLS-SEM employ ML algorithms in their analyses. In the business domain, authors mainly benefit from ML capabilities to validate their PLS-SEM findings and to identify non-linear associations in their models post PLS-SEM (e.g., [Leong et al., 2020](#); [Ray et al., 2021](#)). In addition to these possibilities, researchers can also benefit from identifying new structural relationships in the data using ML algorithms. This approach can further facilitate understanding of influencing factors in business relations, for instance, determining which factors are associated with customer retention, technology adoption, or the performance in multicultural teams, doing so in ways that existing theoretical models cannot (e.g., [Gupta & Kim, 2008](#)).

PLS-SEM and ML share some common ground regarding prediction-related capabilities and metrics, although they follow different philosophies and serve distinct use cases. In prediction, we forecast an output value for new or future observations given relevant input values that are or might be associated with a model's output. To assess the model's accuracy, we assess how well it predicts (output) data that was not used to estimate or optimize the model ([Shmueli, 2010](#); [Shmueli & Koppius, 2011](#)). In PLS-SEM, we build models that combine latent constructs and indicators (or observable variables) and use them in (simultaneously) testing theories and making predictions. The predictions can refer to the indicators, that is, to predict the endogenous constructs' indicator scores (see [Shmueli et al., 2016](#)), or to constructs, that is, to predict the endogenous constructs' construct scores (see [Danks et al., 2017](#)). In PLS-SEM the algorithms and evaluation routines are standardized, which facilitates the interpretation of the results and enables a comparison and selection of SEM modeled by the researcher.

ML uses a diverse range of algorithms to achieve highly accurate predictions. Many of these algorithms are "black boxes" due to their limited transparency in revealing the relationships between (indicator or construct) variables, but not all ML algorithms share this characteristic. Constraint-based algorithms for learning BN allow researchers to identify previously unanticipated relationships that were not actively conceptualized beforehand in pre-specified models. These algorithms introduce a framework of causal reasoning into ML that enables the investigation of causal relationships in the data (e.g., [Druzdzel & Simon, 1993](#)). Through its data-driven causal discovery approach, BN can contribute to uncovering new relationships and refining existing models. These capabilities align well with PLS-SEM's causal-predictive strengths. Combining BN's potential to discover new causal relationships and PLS-SEM's capacity for transparent causal-predictive modeling enables a comprehensive approach that facilitates transparency, model evaluation, comparison, and informed selection of the most appropriate models from competing alternatives (see [Fig. 1](#)).

While researchers call for more combination of methods, such as triangulation or ensembling, to deepen the insights gained through research ([Bruno et al., 2022](#); [Burton & Obel, 2011](#); [Nielsen et al., 2020](#); [Richter et al., 2022](#)), there is limited knowledge transfer between researchers using PLS-SEM and ML algorithms. We facilitate this knowledge exchange and contend that substantial value can be obtained mutually through integrating the strengths inherent in both methodologies. We discuss how specific ML algorithms can complement PLS-SEM and how the two methods can jointly enhance theory and research. This study emphasizes the structural prediction of endogenous constructs for which we propose a routine that complements past approaches. Through an illustrative example, we demonstrate the benefits of this combined use. Ultimately, our study will enable PLS-SEM researchers to integrate specific ML algorithms (BN for causal discovery, and Multivariate Adaptive Regression Splines (MARS) and Generalized Additive Models

(GAM) algorithms that assist in identifying non-linearities) as new tools in their applications. Thus, the study can trigger further contributions to the theoretical and conceptual ideas in the business literature.

2. Approaches to combining PLS-SEM and ML algorithms

2.1. Literature review

To identify previous approaches combining PLS-SEM and ML algorithms in business research, we searched the Web of Science and Scopus databases using a set of keywords¹ and filtered results for business and management outlets between 2016 and today. Having removed duplicates and studies that, upon closer inspection, did not match our search criteria we worked with 37 studies (see [Table 1](#)). These studies employed ML either prior to (in 6 studies) or after (in 30 studies) applying PLS-SEM. One study, [Sener et al. \(2019\)](#), was assigned to both categories. These authors used the universal structural modeling approach that [Buckler and Hennig-Thurau \(2008\)](#) proposed. [Sener et al.'s \(2019\)](#) aim was to reveal previously unknown relationships between constructs (akin to studies characterized as *pre*-PLS-SEM) and non-linear dynamics (akin to studies categorized as *post*-PLS-SEM) through an iterative process using different techniques such as principal component analysis (PCA), ANN, and the conventional PLS-SEM.

The majority of the identified studies used ML algorithms (e.g., ANN) after applying PLS-SEM, mostly to identify non-linearities or to validate their PLS-SEM findings (e.g., [Leong et al., 2020](#), [Lo et al., 2022](#); [Ray et al., 2021](#)). Typically, the studies refer to this method as a hybrid SEM-ANN approach: First, the hypothesized relationships are tested using PLS-SEM, including the standard model evaluation procedures (e.g., [Hair et al., 2019](#)). Second, the exogenous constructs (either all, or only those that show significant relationships to the endogenous construct(s)) are used as input for a subsequent ANN approach. Finally, a sensitivity analysis is used to evaluate the contribution each exogenous construct makes in developing the outcome. Using this procedure, [Leong et al. \(2020\)](#), for instance, identified previously overlooked non-linear associations in consumer decision making processes in social commerce, and [Lo et al. \(2022\)](#) compared the normalized relative importance found in the ANN results to the path coefficients produced by PLS-SEM in consumers' impulsive buying in livestreaming commerce. Only a few studies in business-related research used ML approaches to aid the development of structural models or to explore the relationships between constructs, before applying PLS-SEM. For instance, [Mandhani et al. \(2020\)](#) used a BN represented in a graphical structure to build an exploratory model based on what they learnt from the data. The authors applied the BN to factors obtained from a PCA. They integrated 16 algorithms and analyzed the most robust structure following various criteria.² This structure was then used in the subsequent PLS-SEM analysis.

Notably, most of the above studies were driven by the same set of authors in the field. For instance, all three studies that present a BN before PLS-SEM involve Jyoti Mandhani as a co-author (e.g., [Mandhani et al., 2020](#)); similarly, in the studies that use ML post-PLS-SEM, Keng-Boon Ooi is a co-author 13 times (e.g., [Leong et al., 2020](#); [Ooi et al., 2018](#)). We noted additionally that only two of the reviewed studies ([Lo et al., 2022](#); [Sanchez-Franco et al., 2019](#)) used the PLSpredict routine or its ideas to validate their PLS-SEM's predictive power.

Our approach drew inspiration from these studies, and the discussion of causality in BN (e.g., [Druzdzel & Simon, 1993](#)). It also shows similarity to studies in information and expert systems, transportation, and

¹ Keywords included combinations of PLS-SEM, ML, BN, ANN, in singular and plural format using both abbreviations and the full wording.

² Criteria such as, arc strength and occurrence proportion of arcs from bootstrapping. The structure obtained from the Hill-Climbing algorithm with reference to the AIC metric turned out to be the most robust.

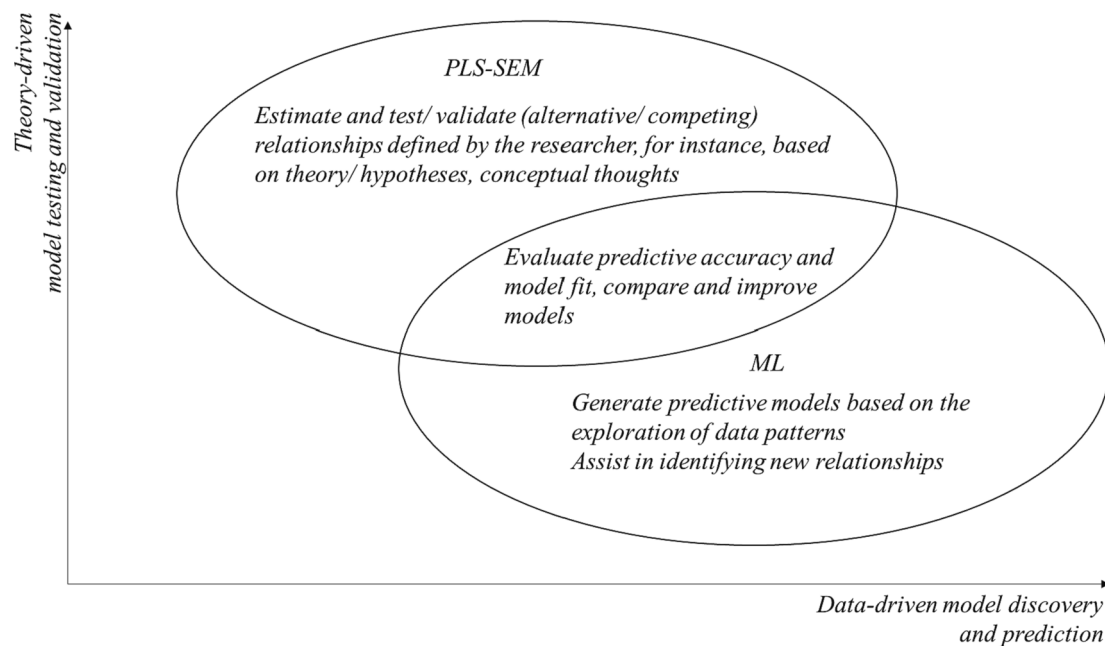


Fig. 1. Theory-driven model testing and data-driven model discovery. *Note: There are ML approaches that can be used for theory testing purposes, for instance, specific forms of BN (see, e.g., Druzdel & Simon, 1993).

Table 1

PLS-SEM and ML: Review of previous approaches in business research.

Approach	N° of articles	Major Purposes (multiple coding enabled)	Main algorithms/ techniques used	Illustrative references
PRE-PLS-SEM: ML before using PLS-SEM	7	Topic/ construct development (N = 3) Model development / exploration (N = 4)	Text-mining, sentiment analysis, data mining BN using directed a-cyclic graphs (DAGs), applying various algorithms (e.g., hill-climbing) (N = 3) ANN	Sanchez-Franco et al., 2019 Mandhani et al., 2020; Sener et al., 2019*
POST-PLS-SEM: ML after using PLS-SEM	31	Identify non-linearities (N = 23) Robustness test/ validation of PLS-SEM findings (N = 8) Capture non-compensatory associations (N = 8)	ANN (N = 26); mostly combined with sensitivity analysis Other approaches: Naive Bayes, decision trees, logistic regression, random forest, support vector machines	Leong et al., 2020; Lo et al., 2022 Loh et al., 2022; Sener et al., 2019*

*[Sener et al. \(2019\)](#) is the only article coded twice as it can be categorized both as pre- and post-PLS-SEM. They use universal structural equation modeling, which was one of the first ML approaches acknowledged in the PLS-SEM community ([Rigdon et al., 2010; Sharma et al., 2021](#)).

the life-science literature that combine SEM and BN techniques to investigate new relationships (e.g., [Carvalho et al., 2021; Díez-Mesa et al., 2018; Gupta & Kim, 2008; Kenett et al., 2022](#)). Finally, our study resembles some aspects of the above studies that used a hybrid SEM-ANN approach to detect non-linearities and validate PLS-SEM results, but we propose a different group of algorithms, as we argue next.

2.2. Selected ML algorithms to complement PLS-SEM

In research using ML, evaluating one or multiple algorithms to identify the most suitable one for a given task is standard practice. A combination of factors determines the algorithm selection, which include its alignment with the research objectives and practical considerations. [Table 2](#) provides an overview of six groups of models that can be used to generate predictions or to investigate causal relationships in the data. The term “model” in ML is not to be confused with the structural model in PLS-SEM; in ML it refers to algorithms that predict or identify patterns in the data.

Business researchers who combined PLS-SEM with ML have predominantly used ANN to validate their PLS-SEM findings and identify non-linearities. We propose a different set of algorithms to achieve these objectives: initially, we implement BN for the exploration of new

relationships in the data, followed by the application of MARS and GAM, which excel at uncovering non-linearities and interactions with greater interpretative clarity (see the bolded assessments [Table 2](#)). We present a more detailed comparison of these algorithms below to provide background information relevant to supporting our algorithm selection.

ANN are inspired by how the brain functions and belong to the class of black-box models. These networks learn and detect patterns in data by adjusting the weights and biases of the connections between their internal nodes (neurons), without explicit programming. ANN’ ability to simulate non-linear interactions using activation functions is one of their key features, allowing them to capture complex patterns in the data. They are beneficial in processing large volumes of data, which makes them suitable for industries like marketing ([Leong et al., 2020; Lo et al., 2022; Ray et al., 2021](#)). ANN allow us to evaluate the predictors’ importance, for instance by using sensitivity analyses, but it is difficult to explain how the model arrives at its predictions ([LeCun et al., 2015](#)). The transparency property is, however, relevant in many business applications, where it is important to understand how the model arrives at its predictions. Additionally, such understanding is relevant to PLS-SEM researchers to support translating the findings into future conceptual models.

BN represent a further group of algorithms that can be beneficial in

Table 2
Overview of selected ML algorithms.

ML algorithms	Linear regression with regularization ¹ White box	MARS and GAM ¹ White box	Regression trees ¹ White box	Bagging, Boosting, Random forests ¹ Black box	ANN ² Black box	BN (Directed Acyclic Graphs, DAGs) ³ White box
Core feature	Control model complexity, prevent overfitting	Identify & explain non-linearities	Graphical model representation & transparency	Control overfitting & increase generalization (stable & robust)	Identify complex non-linearities; large-scale/high-dimensional data; strongest predictive power	Model complexities (multiple causes/ outcomes); handles uncertain/ incomplete information; graphical model representation
Capability to...						
...be used on small, high-dimensional data	High	Low	Low	Low	Low	Low
...determine importance of predictors	Yes	Yes	Yes	Yes	Yes	Yes
...handle complex non-linear relationships	Low	High	Moderate	High	High	Moderate
...enable explanation of non-linearities	Low	High	Moderate	Low	Low	Moderate
...handle complex interactions	Low	High	Moderate	High	High	High
...enable inference of causality from data	Low	Low	Low	Low	Low	High
Further practical consideration						
Non- parametric	No	Yes	Yes	Yes	Yes	Yes
Accuracy	Good	Excellent	Good	Excellent	Excellent	Excellent
Interpreta-bility	High	Moderate	Moderate	Low	Low	High
Training time	Fast	Moderate	Fast	Moderate	Long	Moderate
Transparency of prediction mechanisms	High	Moderate	High	Moderate	Low	High
Recommended software package in R:	glmnet; caret; MASS	earth; caret; mgcv; gam	Rpart; caret	randomForest; gbm	nnet; nlsv; rminer; validann	bnlearn; gRain

Source: ¹ Hastie et al., 2009a; Hastie et al., 2009b; James et al., 2021; ²Goodfellow et al., 2016; ³Pearl, 1988; 2012b.

exploring new structural relationships. Also known as graphical causal models, Directed Acyclic Graphs (DAG), or non-parametric structural equation models (Pearl, 2012a), BN serve for both prediction and causal inferencing (Pearl, 1988). They offer transparency and easy understanding. The graphical structure comprises nodes representing observed or latent variables and (directed) edges or arcs representing variable dependencies. In BN terminology, the node from which an arc directs outwards is a 'parent' node, and the node with an arc directed inwards is a 'child' node. Child nodes are equivalent to endogenous variables, while parent nodes without incoming arcs are considered exogenous variables. This network structure facilitates identifying associations between variables, as well as discovering mediating variables (Neapolitan & Jiang, 2007). While typically employed in discrete variable settings, BN can be adapted to handle continuous variables using a Gaussian distribution to model the probability density function of each variable (e.g., Scutari, 2007). Additional approaches for continuous variables include kernel density estimates, Gaussian mixtures, and copulas for continuous variables (Koller & Friedman, 2009). A distinguishing feature of BN is that both the model structure and the model's parameters can be learned from the data. Learning BN involve determining the graphical structure through algorithms like constraint-based, score-based and hybrid methods. These algorithms search for the structure that best fits the data or they optimize the network structure for prediction. Once the network structure is learned, parameter estimation is performed using techniques such as maximum likelihood and Bayesian estimation. Overall, BN algorithms reveal hidden patterns in the data that might not be immediately evident. Hence, the relationships

between nodes identified by BN offer novel insights for conceptualizing structural models and, potentially, for advancing new theory (Druzdzel & Simon, 1993).

Both MARS and GAM are powerful and flexible ML methods for identifying and understanding non-linear relationships and interactions without the researcher having to manually reshape the data beforehand (Hastie et al., 2009a; James et al., 2021). MARS employs hinge functions to model the relationship between an outcome and a combination of different predictors. This approach can identify important interactions between predictors (Friedman, 1991). GAM offer a complementary understanding of the relationship between each predictor variable and the outcome. In GAM, each predictor-outcome relationship is individually modeled, and the final model aggregates these relationships (James et al., 2021). While GAM are limited to additivity, the valuable interactions unearthed by MARS can enhance GAM' insights. Hence, MARS and GAM are good candidates for enriching the PLS-SEM insights by identifying non-linearities that include interactions and for validating findings in a more transparent way than ANN.

ML algorithms involve hyperparameters³ that are fine-tuned to optimize their performance, influencing how the model learns and generalizes to new data. Unlike parameters, which directly capture data patterns, hyperparameters guide the learning process. Their fine-tuning

³ Some examples of hyperparameters are: the learning rate and the activation function in ANN, the number of hinge functions and the degree of interaction in MARS, the smoothing parameters and the number and placement of knots in GAM (Friedman, 1991; Hastie et al., 2009a; James et al., 2021).

Table 3

Selected relevant metrics and routines to evaluate and choose between models in PLS-SEM and ML.

Metric/ technique/ concept	Definition and interpretation	Typically used in	
		PLS-SEM	ML
Root Mean Squared Error (RMSE)	A metric to assess a model's predictive power; smaller values indicate higher predictive power. Calculated as the square root of the average of the squared differences between the predicted and the actual values. The value depends on the scaling of the (observed or latent) variables used. <u>In PLS-SEM</u>, the model choice relates to evaluating whether the RMSE of the research model is < RMSE of the LM benchmark model (e.g., Shmueli et al., 2019). <u>In ML</u>, there is no universally defined benchmark for the RMSE; the model choice often follows the lowest RMSE or selects the one that is one standard deviation away from the lowest RMSE (to avoid overfitting; James et al., 2021). Similar to PLS-SEM's IA approach (see below), some authors propose comparing the RMSE of the model to the RMSE of a naïve benchmark that predicts the mean or median value of the target variable (Torgo, 2017).	Yes	Yes
Mean Absolute Error (MAE)	A metric to assess a model's predictive power; smaller values indicate higher predictive power. Calculated as the average absolute difference between the predicted and the actual values. It implies treating all errors equally, regardless of their magnitude, and therefore, it is used when all errors are considered equally costly. MAE might be more robust than RMSE in situations where outliers are present. The value depends on the scaling of the (observed or latent) variables used. The insights presented on model choice in <u>PLS-SEM</u> and <u>in ML</u> for RMSE can be likewise applied to the MAE (see above).	Yes	Yes
Bayesian Information Criterion (BIC)	A metric to compare models based on their fit to data and complexity; smaller values generally indicate higher predictive power. Its calculation involves a log-likelihood function, the sum of squared errors, the sample size, and the number of predictors of the construct of interest or the number of parameters in the model; with this, it penalizes complex models and prevents overfitting (MacKay, 2003). It depends on the scaling of observed or latent variables used. <u>In PLS-SEM</u>, it is used to choose between models (e.g., a model with a BIC of -200 is preferred over a model with a BIC of -50). Can be complemented with the BIC-based Akaike weights that provide insight on the model's relative likelihood in percentage; higher percentage values indicate a higher (better) likelihood (e.g., Hair et al., 2022). <u>In ML</u>, smaller BIC values are also better. An exception is the BIC in the <i>bnlearn</i> package: here BIC is subject to a rescaling, so that higher values are better (Scutari, 2007). Larger differences in BIC values favor one model over another, but there is no fixed minimum difference in BIC values that universally indicates whether one model is better than another. Context and domain knowledge matter in BIC-based model selection, therefore using multiple criteria (e.g., k-fold cross-validated RMSE) is advised (Hastie et al., 2009b; Kuhn & Johnson, 2013).	Yes	Yes
Q ² -predict	A benchmark metric to assess a model's predictive power in PLS-SEM. It uses the mean value of the variables in a training sample which acts as prediction for the variables in the holdout sample (called naïve benchmark) and is calculated as one minus the quotient of the PLS path model's sum of the squared prediction errors in relation to the mean value's sum of the squared prediction errors. If Q²-predict > 0, it signifies that the PLS-SEM prediction error is smaller (i.e., better) than that of the naïve benchmark. It supports the model evaluation by evaluating whether the Q²-predict > 0. If this is not the case, predictive relevance is not confirmed. If it is the case, the evaluation continues with the RMSE (if prediction errors are highly symmetrically distributed) or the MAE (Shmueli et al., 2019).	Yes	No
Indicator-averages (IA) benchmark	A benchmark used in the context of predictive model assessment in PLS-SEM. The IA are the indicator averages from the training sample and are used as predicted values, or as a naïve benchmark prediction. The prediction errors of this IA procedure are compared to the prediction errors of the PLS-SEM prediction. Researchers aim for lower prediction errors of their PLS-SEM prediction (Hair et al., 2019 ; Shmueli et al., 2019).	Yes	No
Linear regression model (LM) benchmark	A benchmark used in the context of predictive model assessment in PLS-SEM. A LM generates predictions for the endogenous indicators by running linear regressions with each endogenous indicator as dependent and all exogenous indicators as independent variables. The prediction errors of this LM procedure are compared to the prediction errors of the PLS-SEM prediction. Researchers aim for lower prediction errors of their PLS-SEM prediction (Hair et al., 2019 ; Shmueli et al., 2019).	Yes	No
k-fold cross validation	A technique that enables the predictive model assessment based on holdout-samples that are created by dividing the available dataset into k subsamples or folds. The kth subsample is designated as the holdout-sample, while the remaining k-1 subsamples are used to estimate the model, i.e., as the training sample. This process is repeated k times each time designating a different subsample as the holdout sample. <u>In PLS-SEM</u>, the recommendation is to set k to 10, as long as the minimum sample size requirements in the PLS-SEM context for the training sample are met; the recommendation is to repeat the k-fold cross-validation 10 times, to average out potential abnormal solutions from a partition (see Shmueli et al., 2016; Hair et al., 2019; Shmueli et al., 2019; Lienggaard et al., 2021). <u>In ML</u>, setting k to 10 similarly stands as the most common choice based on several past studies (Kohavi, 1995; Bouckaert, 2003; Refaeilzadeh et al., 2009). In the same way, to ensure a reliable performance estimation the recommendation is to conduct the k-fold cross-validation multiple - often 10 times - to achieve a larger number of estimates, leading to a more robust performance evaluation (Bouckaert, 2003; Refaeilzadeh et al., 2009).	Yes	Yes
Overfitting	Overfitting describes a situation in which the model fits the dataset used very well, but is less able to predict new data, and thereby to generalize to other samples. To avoid this situation, routines that separate training and holdout samples are used. A comparison of the distribution of the prediction errors in the holdout or new sample with the training sample (in-sample cases) can help to diagnose overfitting: a large difference in favor of the training sample prediction indicates potential overfitting (Shmueli et al., 2016). <u>In PLS-SEM</u>, an excessive R² (e.g., ≥ 90 %) similarly indicates a potential model overfit (Hair et al., 2022). <u>In ML</u>, the performance of the model on the training data and the validation data is compared. If a model yields a small training RMSE but a large test RMSE, it indicates overfitting. Also, learning curves with performance trends are evaluated: If the gap between training performance and validation performance widens as more training iterations or examples are added, it indicates overfitting (James et al., 2021).	Yes	Yes

(continued on next page)

Table 3 (continued)

Metric/ technique/ concept	Definition and interpretation	Typically used in	
		PLS-SEM	ML
Loss	A concept of relevance to the CVPAT test in PLS-SEM; a lower loss implies a lower (i.e., better) prediction error. Calculated for each case in the sample as the average squared prediction error over all indicators that are used to measure the endogenous latent variable(s). A model's overall loss is the average of all individual losses, i.e., the losses for each case. In the context of the test, the difference between the average losses of two models in predicting the indicators of the endogenous variables is referred to. The model with the lower loss is superior (Liengaard et al., 2021; Sharma et al., 2023).	Yes	No
Cross-validated predictive ability test (CVPAT)	A test statistic to assess and compare the predictive power of model(s). It enables a statistical significance test that refers to the loss difference between models. It refers to the difference in the average out-of-sample predictive accuracy between two competing models when predicting the indicators of all or specific endogenous construct(s). It can be applied to specific endogenous constructs of interest or to the overall model, i.e., to all endogenous constructs. The model with the smaller loss is preferred. If the loss of Model 1 minus the loss of Model 2 ≤ 0 , then Model 1 is preferred; if the loss of Model 1 minus the loss of Model 2 > 0 and statistically significant (for instance, with $p < 0.05$), then Model 2 is preferred (see Liengaard et al., 2021; Sharma et al., 2023).	Yes	No

Note: This list is not exhaustive; it only refers to metrics/ techniques / concepts discussed in this manuscript.

is essential for balancing model complexity with the ability to fit the training data, ultimately enhancing predictive accuracy.

2.3. Metrics and techniques to evaluate and select models

PLS-SEM shares several features in evaluating models with the broader range of techniques employed in ML (see also Table 3). These encompass metrics to assess the predictive power of models by comparing the predicted and actual values: the root mean squared error (RMSE) and the mean absolute error (MAE), which indicate higher predictive power by smaller values. Also, information-theoretic model selection criteria are used in both PLS-SEM and ML. In particular, the Bayesian information criterion (BIC; Schwarz, 1978) can assist in identifying reasonable models that optimize predictive accuracy and at the same time balance model fit and complexity to avoid overfitting (Grog, 2006; McQuarrie & Tsai, 1998; Shmueli et al., 2019).

The two approaches also distinguish between a training sample (for PLS-SEM estimation or model learning in ML) and a holdout sample (used in validation or cross-validation). Likewise, they use *k*-fold cross-validation to generate multiple holdout samples for improved test metric reliability. This technique divides the data into subsamples (or folds), subsequently using distinct subsamples iteratively for training and testing. To illustrate, Shmueli et al. (2016) developed the PLSpredict routine. This approach estimates the model parameters (e.g., path coefficients, weights, and loadings of indicators) using a training sample. It then applies the model to predict the endogenous constructs' indicator scores. The predicted values for cases in the holdout sample are then compared to the actual values of the indicators in the raw data to evaluate the model's predictive accuracy (Shmueli et al., 2016; Shmueli et al., 2019). If the aim is to predict an endogenous construct score, the routine is adapted because there are no actual endogenous construct scores in the raw data unless estimated. Following Danks et al. (2017) the entire sample is used to estimate composite scores for the endogenous constructs. These scores are called "actual* (actual-star) composite scores" and serve as proxies for the actual scores of the holdout cases to evaluate the model's predictive accuracy.

In addition, the PLS-SEM research community has developed benchmarking routines (e.g., the linear regression model (LM) benchmarks), test statistics, and decision trees (e.g., Shmueli et al., 2019) to evaluate the predictive power of alternative PLS-SEM models (see Table 3). These can aid the selection of competing data-driven or theory-driven models. Notably, to compare two conceptual models' predictive ability, Liengaard et al. (2021) proposed the CVPAT test which was recently extended by the capability to compare two models regarding specific constructs' prediction accuracy (Sharma et al., 2023). To select a suitable model, authors propose first comparing the BIC of the alternative models. If the BIC differences between models are however small,

researchers are advised to compare models using the CVPAT statistical test (Hair et al., 2022). Nonetheless, particularly if the models to be compared have resulted from an exploratory process, the alignment of model selection with theoretical or conceptual considerations remains crucial.

In PLS-SEM, the model selection process remains confined to constructing and evaluating a restricted array of alternative models or theories. The selection is inevitably influenced by the researcher's perspective and assumptions. Considering this limitation, we recommend leveraging PLS-SEM's predictive capabilities in joint ML and PLS-SEM approaches.

3. A proposal to combine PLS-SEM and selected ML algorithms in future studies

We recommend transferring insights from the BN, MARS, and GAM algorithms into PLS-SEM. Specifically, we propose a routine to explore new relationships between existing constructs, improve existing conceptual models, and to compare an existing model with one or more improved ones (see Fig. 2).

3.1. Step 1: Create and estimate a baseline model in PLS-SEM including PLSpredict

The first step is to create a PLS-SEM on an initial conceptual model involving baseline theorizing. For this model, the researcher needs to prepare and check the data following the PLS-SEM standards, considering threshold values for more exploratory purposes (e.g., Hair et al., 2022). This model can form the baseline benchmark the researcher aims to improve by combining PLS-SEM and ML approaches.

We propose extracting the latent variables scores from PLS-SEM to be used in ML models. These scores can be based on the initial conceptual model. Generating these composite or latent variable scores in PLS-SEM considers the structural model's context, that is, the scores represent the underlying modeled theory (Hair et al., 2017; Rigdon et al., 2019). Using the initial theoretical model to extract latent variable scores, therefore, implicitly involves the relationships modeled. This strategy has been implemented by most researchers who combined ML and PLS-SEM in the past and could be one way forward. Another way forward that considers the overall context of constructs while avoiding creating latent variables scores based on pre-specified relationships, is to create a path model that connects all exogenous latent variables involved in the initial theoretical model with one another and with the key exogenous target construct to be predicted. Thereby we consider the full context of constructs. The latent variable scores are then estimated using the factor weighting scheme, which, unlike the path weighing scheme, does not differentiate different directions of relationships between constructs (e.g., Sarstedt

et al., 2022c). This could be advantageous to avoid a biased assessment of the model structure and performance when the ML approaches are being implemented.

3.2. Step 2: Apply selected ML algorithms on the latent variable scores extracted

3.2.1. Explore the model structure using BN algorithms

For the context of this study, the retrieved scores can be employed in the ML framework for one of two objectives. First, they can be used in BN to discover existing and non-existing associations and mediating effects among variables. BN operationalize causal assumptions by aligning them with observable data, delineating variables' associations, and statistical independencies (Elwert, 2013). The architecture of BN allows for adding "a priori" knowledge, which can be combined with data exploration efforts. This pre-existing knowledge, particularly regarding the presence or non-existence of a relationship between variables (i.e., the "structure prior") can aid model structure learning. In searching for the best BN structure and therewith for alternative models, the researcher can use two main approaches: constraint-based and score-based algorithms.

Constraint-based algorithms operate through a series of constraints to infer the conditional independence relationships between variables. These constraints are used to search for the optimal structure that explains the observed data, and thereby forms the final BN structure (Spirtes et al., 2000). A key step in the process is the independence test. It determines whether, given a set of variables, two selected variables are conditionally independent. This procedure involves comparing the observed data against the distribution expected if the two variables were independent. The goal is to construct a structure that exclusively encompasses relationships existing within the actual data. One of the most well-known algorithms used in BN is the Grow-Shrink (GS) algorithm (see a discussion in: Neapolitan, 2003; Neapolitan & Jiang, 2007). Other common constraint-based algorithms include the Incremental Association (Tsamardinos et al., 2003), and the Fast Incremental Association algorithm (Yaramakala & Margaritis, 2005). While these algorithms yield similar structures, we chose to report and recommend the GS algorithm (Margaritis, 2003) due to its established presence in the literature and its associated accumulated knowledge (Srivastava et al., 2023). It starts with an empty graph and iteratively adds or removes relationships based on a test criterion (like the conditional independence tests, i.e., partial correlation tests⁴) and the level of significance set by the analyst. It focuses on finding the overall structure of the network and offers information on relationships between variables, including the direction (though sometimes just reporting an association without a direction). Its limits are primarily in its background assumptions involving the non-existence of unmeasured common causes in the data and the parametric assumption that the causal dependencies can be modeled by linear functions (Rau & Scheines, 2012).

Score-based algorithms find the structure that best fits the data, based on a scoring function (e.g., BIC) that measures the quality of the fit (Scutari & Denis, 2021). By finding the structure that optimizes the score, the algorithm can identify the relationships between the variables that the data most supports. The Hill-Climbing (HC) algorithm is a well-known score-based algorithm. It is a greedy algorithm that adds one relationship at a time based on local optima (Neapolitan, 2003). Due to the greedy nature of this search approach, the model building process is repeated numerous times with alternative initialization settings. The algorithm automatically chooses the model with the best score (i.e., the best BIC; Neapolitan, 2003).

⁴ Voortman and Druzdzel (2008) demonstrated experimentally that the partial correlation test is very robust against departures from normality, and thus, yields robust results. Only if the deviation from multivariate normality is very large, the test might yield incorrect results.

Both the GS and the HC algorithms offer graphical representations of the model, illustrating directed relationships between the variables (or constructs) with arrows and non-directed relationships with lines. These visualizations assist in validating established relationships, highlighting non-existing relationships, and identifying new relationships, including mediation effects. The algorithms also generate cross-validated RMSE and BIC values which, moreover, can assist us in understanding potential improvements on the benchmark model in PLS-SEM.

3.2.2. Identify non-linearities and interactions using MARS and GAM

Subsequently, these scores (and models found) can be used to discover non-linear relationships and interaction effects, for which we recommend the use of MARS and GAM (Hastie et al., 2009a; James et al., 2021). MARS allows for representing non-linear relationships through a piecewise linear approach, effectively decomposing complex relationships into a succession of linear segments. To specify, MARS recursively partitions the predictor space into smaller regions and fits a linear regression model into each region called "hinge functions". The position and shape of the hinge function reveal the specific input space region where a predictor has the most impact on the target variable. A significant positive coefficient associated with a hinge function, such as $h(x-a)$, indicates a non-linear relationship, with "x" as the input variable and "a" as the threshold value that marks the boundary between different regions of the input space. The hinge functions can also identify interaction effects between predictors, for example, when the input space is split based on the values of two predictors. However, interpreting the results should be done cautiously because the non-linearities found with MARS might not necessarily hold in PLS-SEM. This discrepancy can arise, for instance, because significant interactions that are found may only apply to a specific range and not the full range of input values of the variables involved.

GAM enhance the modeling of complex relationships by partitioning the predictor space and applying polynomial basis functions in each partition, for increased modeling flexibility. To assess non-linearity between predictors and outcomes, researchers typically examine the estimated effective degrees of freedom (edf) from GAM (Wood, 2006). The edf values provide insight into the extent of non-linearity in the relationship: an edf of 1 implies a linear association, an edf between 1 and 2 indicates mild non-linearity, and an edf above 2 represents substantial non-linearity. This interpretation is based on the idea that the edf reflects the number of knots - junction points of the piecewise polynomial components - in the model fitting (Wood, 2006). Conventionally, a model selection process involves fitting GAM with progressively fewer knots, seeking a parsimonious model that retains the relationship's essential features, as determined by graphical inspection. This stepwise reduction in knots limits the complexity of the model, yielding a more conservative edf estimation. Using the *mgcv* package in R (Wood, 2022) researchers can automatically determine the optimal level of non-linearity (number of knots) for GAM models through cross-validation, streamlining the model fitting process (Wood, 2022).

Although MARS and GAM are not designed to test variable networks or structural models, they offer valuable insights through partial dependency plots with confidence intervals, enabling us to understand the form of the relationships better. Additionally, an analysis of deviance (James et al., 2021) can be used to determine which of the models is best. If the non-linear model has a significantly lower deviance than the linear model, it suggests a better fit to the data. Further, criteria such as the RMSE, the MAE, and the BIC values can help assess the significance of the non-linear relationships compared to alternative models.

3.3. Step 3: Assess the theoretical plausibility of models and relationships identified

Determining causal relationships from observational data alone is challenging (Bollen & Pearl, 2013; Pearl, 2000). Researchers are advised to evaluate the promising models following theoretical considerations,

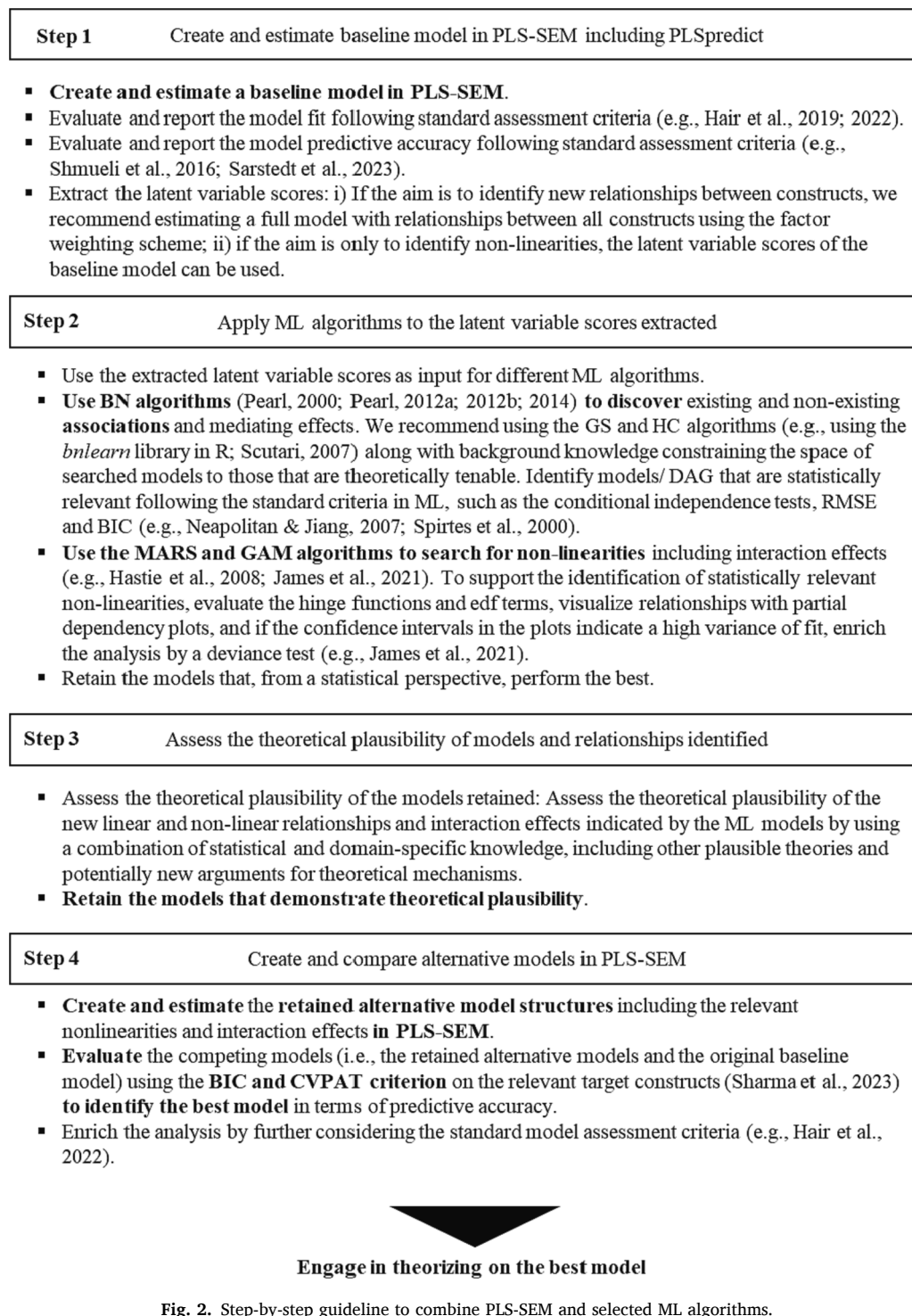


Fig. 2. Step-by-step guideline to combine PLS-SEM and selected ML algorithms.

that is, they need to assess whether there is a causal explanation that could underlie relationships. Hence, after having identified a set of well-performing alternative models using the BN algorithms, the researcher would be well advised to evaluate the theoretical plausibility of these models and relationships (asking whether the indicated new linear relationships are consistent with theory or plausible theoretical mechanisms). Similarly, the researcher should determine whether there is a theoretical logic that can undermine a potential interaction effect indicated in MARS. And finally, researchers should ask about a possible reason for hypothesizing a non-linear mechanism that fits the form indicated by the MARS and GAM models.

In this step, the researcher might need to identify potentially alternative theories and engage in theory development, if the relationships indicated offer room for new causal mechanisms to be outlined between constructs. We recommend engaging in this theoretical plausibility test early to avoid further development leading in a confounding direction. For instance, if the researcher has tested for alternative linear models, the theoretical plausibility assessment may precede a further test for non-linearities. Otherwise, the further implementation and testing of models in PLS-SEM could become quite complex and might involve models that will not be useful for later theory development.

3.4. Step 4: Create and compare alternative models identified in PLS-SEM

In step 4, going beyond previous approaches, we recommend modeling the new structures, interactions and non-linearities (that passed the theoretical plausibility test) in PLS-SEM. Modeling these relationships allows researchers to profit from the interpretability of PLS-SEM findings and enables them to apply recent test procedures in the PLS-SEM context, especially CVPAT (Liengaard et al., 2021; Sharma et al., 2023) to compare alternative or competing models. Thus, the researchers can use the well-established quality criteria in PLS-SEM, both predictive and explanatory, to assess the baseline conceptual model against a series of competing theoretically and data-driven plausible models.

Similar to modeling potentially new linear relationships, the modeling of indicated interaction effects is straightforward (see also Becker et al., 2023). In addition, the researcher can represent non-linearities by exponentiating the input values and then apply a linear model to these values. The non-linear effects can be modeled in PLS-SEM as higher-order terms (typically quadratic) based on the visual inspection of shapes of the partial dependency plots in MARS and GAM. Quadratic (progressive and degressive) relationships are still easy to interpret and theorize. The higher the degree of the polynomial, the more complex the theorizing, though in some fields, this is common, and researchers hypothesize, for instance, on S-curve effects (e.g., Contractor et al., 2003; Richter, 2007).

Having created all alternative models in PLS-SEM, we follow the routine given in Sharma et al. (2023) and compare the proposed models' RMSE or MAE with the naïve prediction benchmark (i.e., IA and LM) to ensure that they meet the minimum predictive accuracy requirements. If these requirements are met, we compare the models by focusing on the BIC and the key target constructs' prediction accuracy using the CVPAT test on the construct level (see also Hair et al., 2022). This assessment offers the input for the researcher to identify the best model and engage in further theorizing. Additionally, researchers can complement the analysis by evaluating the quality criteria in ML compared to the quality criteria reported in PLS-SEM (e.g., the RMSE and BIC used in both approaches). Better performance metrics generated in the ML environment could point to complex model structures and relationships that the researcher might not be able to represent accurately in PLS-SEM (e.g., due to the constraints in modeling extreme non-linearities).

4. Illustrative example: UTAUT

To illustrate the combined use of PLS-SEM and our selected ML algorithms, we used the Unified Theory of Acceptance and Use of Technology (UTAUT) model by Venkatesh et al. (2003) and the data provided by Al-Gahtani et al. (2007). The dataset which contains 722 observations and 21 indicators (see Appendix 1), is available as an example in SmartPLS 4 (Ringle et al., 2022). In its original form, it involves two endogenous constructs, the behavioral intention (BI) and the use behavior (USE), as well as four exogenous constructs, performance expectancy (PE), effort expectancy (EE), social influence (SN), and facilitating conditions (FC). The conceptual model is given in Fig. 3, left-hand side.

Cho et al. (2022a) used this model to illustrate a prediction-orientated specification search algorithm that, similarly to our approach, has the objective of revealing structural models that maximize predictive accuracy. Specifically, the algorithm searches for a structural model that minimizes the target variables prediction errors based on generalized structured component analyses (GSCA). The authors identify an improved version of the original UTAUT that performs better in terms of prediction metrics and is theoretically reasonable. This model is also given in Fig. 3, right-hand side. We refer to both models for benchmarking the performance of the ML approaches presented.

4.1. Step 1: Create and estimate a PLS-SEM including PLSpredict

First, we estimated the original model using the PLS algorithm with SmartPLS 4 (Ringle et al., 2022). As the large majority of criteria meet the required thresholds, we find the measurement models' quality satisfactory (see Appendix 2). We do not discuss these further since evaluating measurement models has become a well-documented standard routine (e.g., Hair et al., 2022; Sarstedt et al., 2023). Having ensured sufficient quality of measurement models, we turn to the structural model results provided in Table 4 and to the results of the PLSpredict routine using k-folds cross-validation presented in Table 5.

The VIFs are all clearly below the critical level of 5 (see Becker et al., 2015), which indicates that multicollinearity is no problem in the structural model. This model satisfactorily explains BI and USE with moderate R^2 values of 0.353 and 0.250 (Hair et al., 2019). Also, the modeled relationships receive empirical support: all path coefficients are significant ($p < 0.05$), and the strongest associations are found for BI on USE, and EE on BI. The PLSpredict routine demonstrates that the model does have predictive validity in that the Q^2 -predict values are above zero and the PLS-SEM predictions outperform the IA benchmarks; however, the PLS-SEM predictions' RMSE and MAE values are higher than those of the LM benchmark for all BI and USE indicators (except USE2) and overall (Shmueli et al., 2019; Sharma et al., 2023). The RMSE of the BI construct indicators vary between 1.052 and 1.243, the RMSE of the USE construct vary between 0.069 and 0.085. The RMSE for the endogenous constructs is 0.815 for BI and 0.937 for USE. Hence, in terms of its predictive accuracy, this model could be improved, which makes it a good fit for our illustrative purposes.

To generate latent variable scores to be used in the ML setting, we estimated a full model that connects all constructs using the *factor weighting scheme* (see Appendix 3). Thereby, we consider the full context of constructs, but do not imply a specific theoretical model in estimating the scores. To understand the difference between the full model latent variables scores and the latent variables scores derived from the PLS algorithm on the original UTAUT, we evaluated the correlations between the two kinds of latent variable scores. We found the lowest correlation for FC with a correlation coefficient of 0.957 ($p < 0.001$); the latent variable scores of EE, PE, SN, and USE are perfectly correlated, i. e., they have a correlation coefficient of one. Additionally, we analyzed the score differences, which point to some rather minor disparities (see Appendix 4).

4.2. Step 2: Apply selected ML algorithms on the latent variable scores extracted

4.2.1. Explore the model structure using BN algorithms

In the second step, we applied a BN for causal discovery that could assist in validating the model's causal structure. To identify potential new causal relationships including mediating effects, we used the latent variable scores extracted from PLS-SEM as input data for learning the BN. We implemented a constraint-based algorithm, GS, and a score-based algorithm, HC, using the software R and the package *bnlearn* (Scutari, 2007). We undertook a structure search that relied on prior information, which allowed us to include (whitelist) or exclude (blacklist) certain relationships in the learning process. Blacklisting constrains the algorithm's search space by excluding relationships that conceptually are unlikely or impossible to occur. In our example, we blacklisted the relationships from the supposed endogenous constructs USE and BI to the exogenous predictors (PE, EE, SN, FC) based on the rich technology adoption research that considered the determinants of intention and behavior (e.g., Blut et al., 2022).

Fig. 4 on the left-hand side shows the model structure resulting from the GS algorithm. It identifies significant relationships (5 % threshold) that have a direction from one construct to the other (arrows) and conditional dependencies (lines). The identified structure resembles the original model, with BI as a mediating construct. Additionally, it

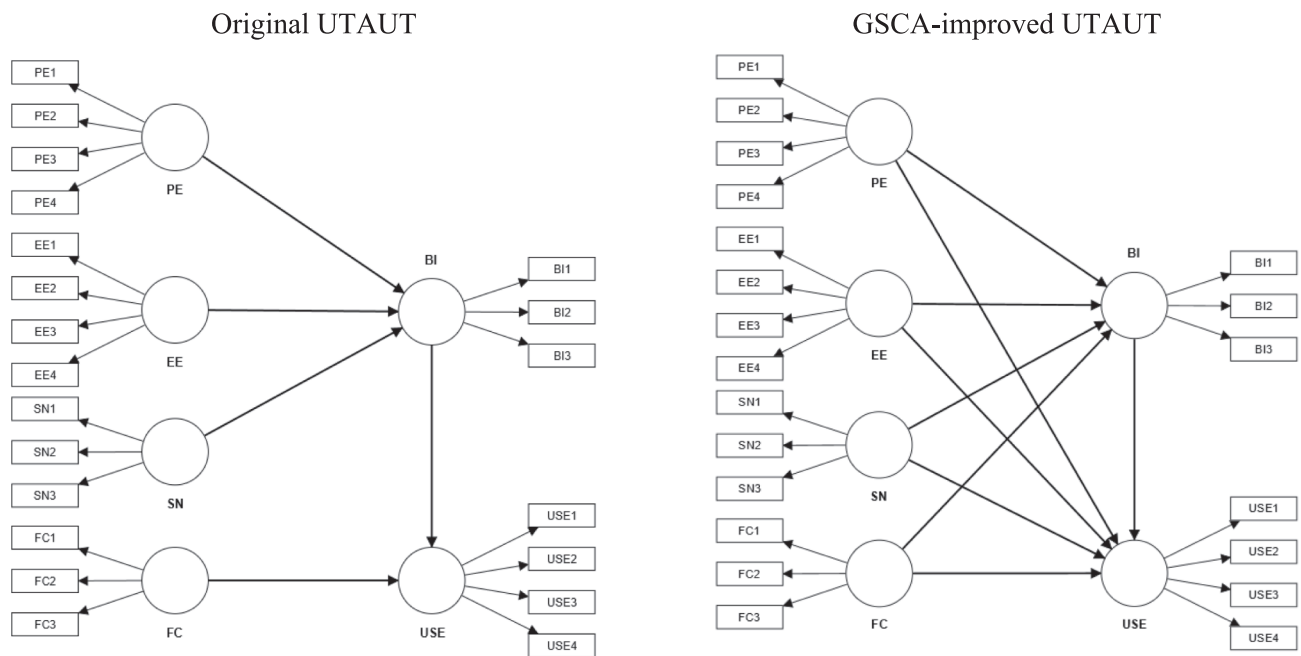


Fig. 3. The original UTAUT (left) and the GSCA-improved UTAUT (right) in PLS-SEM. Source: original UTAUT, left, Venkatesh et al., 2003; GSCA-improved UTAUT, right, Cho et al., 2022a.

Table 4

PLS-SEM results structural model: Original UTAUT model.

Path	VIF < 5	Path coefficient	95 % CI	Significant ($p < 0.05$)?	f^2 effect size
PE → BI	1.307	0.215	[0.134; 0.298]	Yes	0.054
EE → BI	1.243	0.362	[0.292; 0.432]	Yes	0.163
SN → BI	1.133	0.208	[0.135; 0.278]	Yes	0.059
FC → USE	1.284	0.198	[0.126; 0.276]	Yes	0.041
BI → USE	1.284	0.375	[0.285; 0.458]	Yes	0.146
R^2 BI			0.353		
R^2_{adjusted} BI			0.350		
BIC BI			-288.775		
R^2 USE			0.250		
R^2_{adjusted} USE			0.248		
BIC USE			-189.367		

Note: Total effects: PE → USE: 0.081 ($p < 0.05$); EE → USE: 0.136 ($p < 0.05$); SN → USE: 0.078 ($p < 0.05$).

We used the percentile bootstrapping procedure with 5,000 subsamples.

includes one new directed relationship from FC to BI. Hence, the learned structure confirmed all the relationships proposed in the original model and discovered an additional directed relationship. Also, it pinpointed some associations without an explicit direction reflecting an interdependency between constructs (the associations between EE and FC, EE

and PE, and PE and SN).

On the right-hand side, Fig. 4 visualizes the model structure indicated by the HC algorithm that optimizes the model's goodness of fit to the data with a penalty for the number of parameters in the structural model (using the BIC). The algorithm stops when no more relationships

Table 5

PLSpredict results: Original UTAUT model.

Indicator	$Q^2_{\text{predict}} > 0$	PLS-SEM RMSE	LM RMSE	PLS-SEM – LM RMSE < 0	PLS-SEM loss – IA benchmark loss < 0	PLS-SEM loss – LM benchmark loss < 0
BI1	0.187	1.052	1.031	0.021		
BI2	0.049	1.243	1.213	0.030		
BI3	0.291	1.169	1.153	0.016		
USE1	0.074	1.485	1.484	0.001		
USE2	0.069	1.166	1.169	-0.003		
USE3	0.085	1.298	1.277	0.021		
USE4	0.073	2.262	2.228	0.034		
BI	0.341	0.815			-0.299**	0.051*
USE	0.129	0.937			-0.209**	0.050
Overall					-0.247**	0.051*

Note: * $p < 0.05$; ** $p < 0.01$.

can be added without improving the BIC score (see Table 3, and the specifics for the BIC interpretation when using *bnlearn*: higher values are better). We applied bootstrapping to our data which involved learning a set of 500 (network) structures and building the average DAG/structure by setting a threshold (between 0 and 1). The threshold represents the minimum proportion of runs in which a particular relationship must appear for it to be included in the final average network. We identified an optimal threshold of 0.798, meaning that only relationships that appear in 79.8 % or more of the bootstrapping runs were included in the final model (Scutari & Nagarajan, 2013). The resulting structure was similar to the previous ones. Compared to the original model, there is one additional relationship between FC and BI (as also indicated by the GS algorithm), two additional relationships between FC, SN, and USE, and three further relationships between the exogenous constructs, namely from EE to PE, from EE to FC, and from PE to SN.

Given its optimization for BIC, the HC algorithm showed a slightly superior performance in the BIC score compared to the GS algorithm, with only a marginal difference observed between the two models' BIC scores. Evaluating the cross-validated RMSE, the GS and HC performed very similarly in predicting the two outcome variables BI and USE. Hence, both models are interesting alternative models to be evaluated from a theoretical perspective.

4.2.2. Identify non-linearities and interactions using MARS and GAM

To identify non-linearities and interactions between variables, we applied MARS and GAM. We referred to the GS model given in the previous section (Fig. 4, left).⁵ Hence, we used the following equations: $BI = f(PE, EE, SN, FC)$ and $USE = f(BI, FC)$. We fitted MARS models using the *earth* function integrated in the library *caret* in R (Kuhn, 2008). For model training, we combined the number of hinge functions and the degree of interaction between the hinge functions to identify the optimal combination of these two hyperparameters that minimize the cross-validated RMSE. The total number of hinge functions is based on the number of variables and the threshold values that mark the boundary between different regions of the input space. During training, the algorithm applies a backward elimination of hinge functions (or terms) by looking at change in the generalized cross-validated (GCV) R^2 as each new term is added to the model. A term is not retained if its inclusion results in less than 0.001 improvement in the GCV R^2 . In addition to tuning the number of hinge functions, we assessed potential interactions between different terms - using a maximum 3rd degree interaction as recommended in the literature (Boehmke & Greenwell, 2019). Table 6 presents the MARS model summary, as well as the GCV R^2 , RMSE, and MAE.

For BI (left), several non-intercept terms were retained, which were based on the four predictors (PE, EE, SN, FC) used to make the hinge terms. Similarly for USE (right), several non-intercept terms and two term-interactions, based on the two predictors BI and FC, were retained. For both models, the results suggest several non-linearities and an interaction effect ($FC * BI$) within specific input value ranges. MARS allows an estimation of the variables' importance (how much each variable contributes to the model's prediction performance) based on the impact on the GCV error as each predictor is added to the model. Table 6 reveals that EE, FC, and PE are the most important BI predictors and that BI is the most important USE predictor.

As a complementary approach to MARS models, we also fitted GAM (Wood, 2006) for validation and further insight. The edf based on GAM (see Table 7) indicates that for SN and FC, degree 1 and, therefore, a linear relationship is sufficient. For PE, EE, and BI, an edf > 3.440, 2.697, and 1.795, respectively, suggest possible non-linearities to capture the underlying patterns in the data. Fig. 5 presents the relationships we found between BI and EE, PE, SN, and FC, and between USE and BI

and FC in partial dependency plots (PDP). The plots visualize the non-linearities indicated for BI on USE (edf > 1), and for PE and EE on BI (edf > 2). The remaining relationships are defined by a single polynomial of degree 1, which are represented as straight lines.

The plots also demonstrate considerable variability in the fit at the left outer ranges of the PE, EE, and BI predictors (indicated by the dotted lines outlining the 95 % confidence intervals). Notably, only 19 training observations fall in the category where $PE < -5$, $EE < -3$, and $BI < -3$, which results in a high standard error. Therefore, we used an additional test to assess the difference in deviance between a simple model incorporating a linear predictor function and the more complex model integrating non-linear elements. The aim was to determine whether the non-linear model deviance had been significantly reduced. Such a finding would imply that the non-linear model aligns better with the data than the linear model. Table 8 reports the results of this test. For BI, Model 1 is a GAM that uses linear functions for all predictors, and Model 2 is a GAM that uses non-linearities of degree 3.440 for PE and 2.679 for EE. The significant test results indicate that Model 2 which includes the non-linearities, is preferred. Similarly, for USE, Model 1 represents the linear GAM, and Model 2 the GAM including the non-linearity we found earlier. Here, the linear model (Model 1) is preferred, as the non-linear function of BI does not decrease the deviance significantly ($p > 0.05$).

Overall, the GAM applied to the GS model indicate non-linear relationships for PE and EE. From visual inspection of the plots, the relationship between EE and BI seems optimally characterized by a quadratic function and a cubic function might be the best to represent the relationship between PE and BI. Additionally, the MARS models suggest an interaction effect between BI and FC in influencing USE.

4.3. Step 3: Assess the theoretical plausibility of models and relationships identified

Before initiating the next step, we needed to evaluate the theoretical plausibility of the explored relationships. A remarkable variety of articles has created different versions and extensions of the UTAUT building on different theoretical arguments. Blut et al. (2022) provide a synthesis in their meta-analysis. Based on this review, there are theoretical arguments for a relationship between FC and BI as indicated by both the GS and HC algorithms. Conceptual thoughts building on the Theory of Planned Behavior indicate that FC can act similarly to perceived behavioral control and can thereby influence intention (Ajzen, 1991; Venkatesh et al., 2012). Additionally, there are alternative ideas to support the relationship between SN and USE. A first line of argument is related to the high correlation between habit (for which there is a positive association with USE) and SN (as given in Blut et al., 2022). In models that do not include habit, SN may absorb some of the direct impact habit has on USE (e.g., Cho et al., 2022a). A second line of argument is related to coercive pressure, which can potentially explain the direct relationship between SN and USE (e.g., Bozan et al., 2016). Thereby the relationships the HC algorithm indicates between SN and USE have some backing. However, we encountered challenges when seeking to substantiate the plausibility of all the relationships found between (initially exogenous) constructs in the HC model (e.g., between FC and SN). Hence, we suggest further tests of the more parsimonious GS model from a theoretical perspective.⁶

Accordingly, considering the non-linearities and interactions indicated by MARS and GAM, we noted that Blut et al. (2022) recommend assessing interactions between the UTAUT's predictors, as well as the curvilinear effects that could, for instance, be based on optimal stimulation level theory (Mittelstaedt et al., 1976). The interaction effect between BI and FC seemed intuitive, as facilitating conditions can reinforce an intention for use. The non-linearities visualized earlier did

⁵ We refer to the GS model for illustrative purposes here, as we believe it is conceptually more intuitive than the HC model – see our argument later.

⁶ We note that other conclusions can be drawn; the illustrated approach would be the same for the HC algorithm.

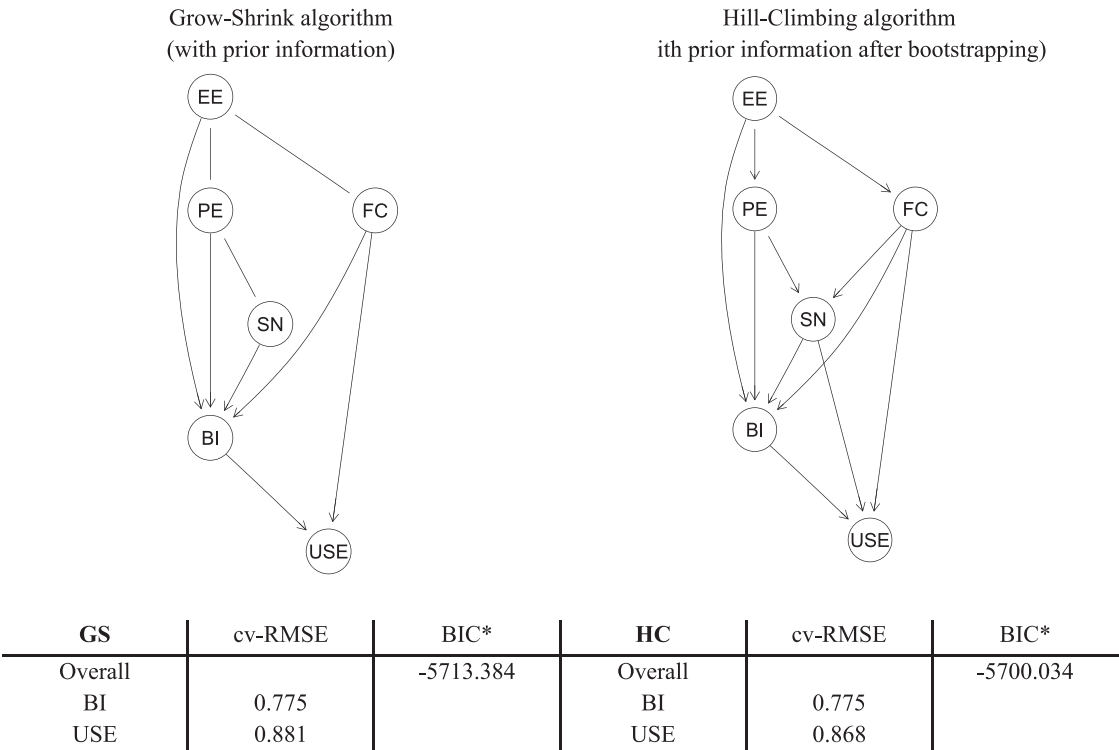


Fig. 4. Structures identified by the GS algorithm (left) and the HC algorithm (right). *Note: BIC calculations follow the *bnlearn* package, i.e., higher values are better. See Table 3 for details.

Table 6
MARS results on the GS model: Overview of coefficients and quality criteria.

Model and predictors BI = f (PE, EE, SN, FC)	CV coefficients after pruning	Model and predictors USE = f (BI, FC)	CV coefficients after pruning
Intercept	−1.525	Intercept	0.487
h(EE- −1.41469)	0.309	h(1.1381-FC)	−0.289
h(−1.78202 - PE)	−0.475	h(BI- −0.393198)	−3.239
h(FC- −2.04857)	0.260	h(BI- −0.172881)	8.531
h(SN- −2.04081)	0.200	h(BI- −0.00440335)	−13.275
h(PE- −0.374069)	0.233	h(0.100529-BI)	−0.604
		h(BI-0.100529)	9.652
		h(BI-0.747055)	−11.605
		h(BI-0.804747)	11.239
		h(1.1381-FC) * h(0.478871-BI)	0.098
		h(FC-1.1381) * h(0.100529-BI)	4.010
R ²	0.404	R ²	0.877
RMSE	0.776	RMSE	0.244
MAE	0.589	MAE	0.693

Note: The variable relative importance for BI = f (PE, EE, SN, FC) is: EE (100), FC (46.3), PE (22.0), SN (<1.0). Variable importance for USE = f (BI, FC) is: BI (100), FC (<1.0). The first predictor with a value of 100 means that it has the highest relative importance of all predictors in the model.

Table 7
GAM results on the GS model: edf values, significance and quality criteria.

Model and predictors BI = f (PE, EE, SN, FC)	edf	F-test	Model and predictors USE = f (BI, FC)	edf	F-test
s(PE)	3.440	10.47**			
s(EE)	2.679	17.51**			
s(SN)	1.000	37.89**			
s(FC)	1.000	53.50**	s(FC)	1.000	32.99**
			s(BI)	1.795	38.32**
R ²	0.400		R ²	0.227	
RMSE	0.753		RMSE	0.870	
MAE	0.575		MAE	0.687	

Note: **p* < 0.05; ***p* < 0.01.

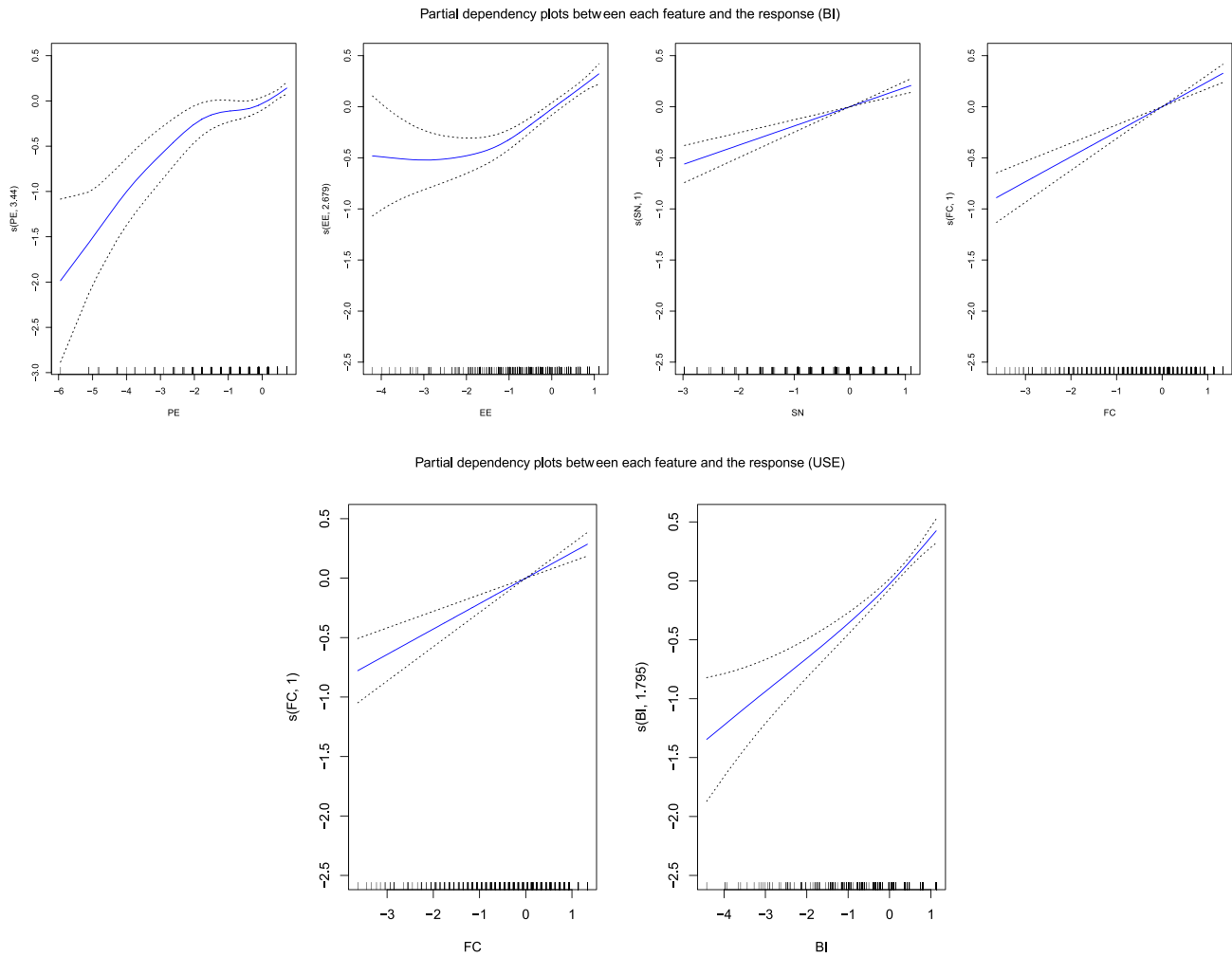


Fig. 5. Partial dependency plots visualizing relationships found (including confidence intervals).

Table 8
Analysis of deviance.

Model	Resid. df	Resid. Dev (RSS)	df	Deviance	F
BI = f (PE, EE, SN, FC)					
Model 1:	716	426.28			
BI ~ s(PE, 1) + s(EE, 1) + s(SN, 1) + s(FC, 1)					
Model 2:	712	418.12	4.109	8.166	3.387**
BI ~ s(PE, 3.440) + s(EE, 2.679) + s(SN, 1) + s(FC, 1)					
USE = f (FC, BI)					
Model 1:	719	552.32			
USE ~ s(FC, 1) + s(BI, 1)					
Model 2:	718	550.31	0.795	2.014	3.306
USE ~ s(FC, 1) + s(BI, 1.795)					

Note: * $p < 0.05$; ** $p < 0.01$.

not fully fit the ideas of an optimal level of stimulation; rather, they pointed to EE having a progressive effect on the behavioral intention and PE having an s-curve effect on BI. Hence, their theoretical plausibility could be debatable, and researchers might refrain from further testing these. We decided, for illustrative purposes, to offer the results of the further testing in PLS-SEM.

4.4. Step 4: Create and compare alternative models identified in PLS-SEM

In the final step, we transferred the new models identified into PLS-SEM and used PLS-SEM's capabilities to compare them to the original UTAUT model as well as to the GSCA-improved UTAUT. Hence, we created the following alternative models in PLS-SEM (see Fig. 6):

Alternative Model (AM) 1 is the improved UTAUT using the GS algorithm, AM2 is the improved UTAUT using the HC algorithm, AM3 is the improved UTAUT using the GS algorithm including the identified interaction effect of BI and FC on USE, and finally, for illustrative purposes, AM4 is the improved UTAUT using the GS algorithm including a cubic non-linear effect of PE on BI and a quadratic non-linear effect of EE on BI. We ran the PLS and PLSpredict algorithms on all models. In addition, we used the CVPAT procedure to compare the models along their predictive power (Sharma et al., 2023).

All models performed better than the IA benchmark, and none performed better than the LM benchmark, which means all models have predictive validity. Yet, as shown in Table 9, it is not strong predictive validity. We found the lowest RMSE for BI in the HC model (0.872) and

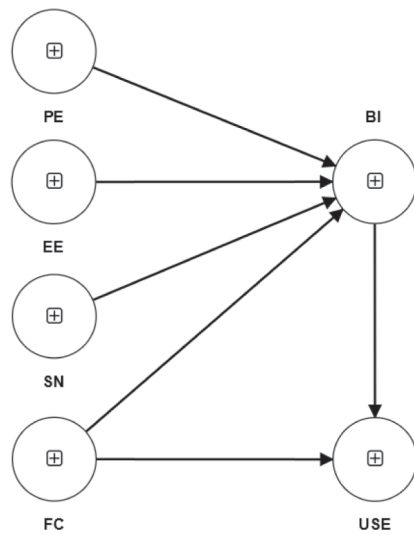
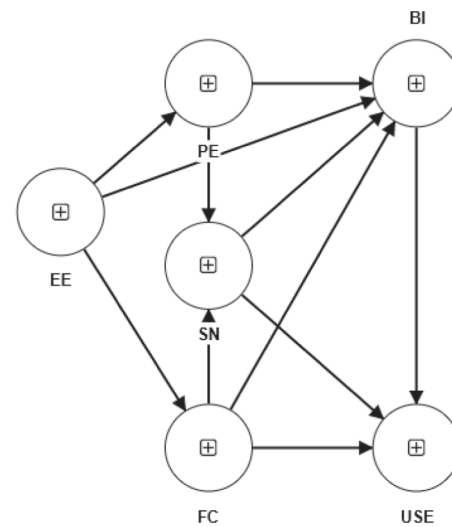
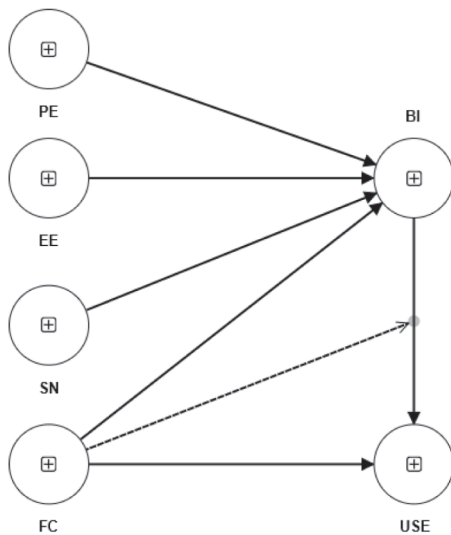
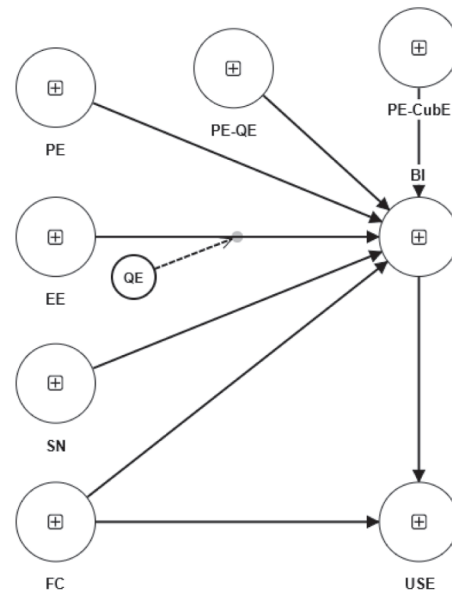
Alternative Model 1: ML-improved UTAUT,
GS algorithmAlternative Model 2: ML-improved UTAUT,
HC algorithmAlternative Model 3: ML-improved UTAUT,
GS algorithm including interactionAlternative Model 4: ML-improved UTAUT,
GS algorithm including non-linearities

Fig. 6. The alternative models in PLS-SEM.

the lowest RMSE for USE in the GSCA-improved UTAUT (as proposed by [Cho et al., 2022a](#)). To compare and select models, the recommendation is that researchers compare the BIC scores. We found the lowest BIC for BI, with a score of -329.941 , for the improved UTAUT using the GS algorithm, and we found the lowest BIC for USE, with a score of -211.418 for the improved UTAUT using the HC algorithm. As some of the BIC values were rather close, we complemented this procedure with the CVPAT test. Hence, we finally compared the models by focusing on the BI's and USE's prediction accuracy using the CVPAT test on the construct level for BI and USE (see [Table 10](#)).

We tested the alternative (linear) models AM1 and AM2 that stem from the GS and HC algorithms against the original UTAUT. The results indicate that both linear models identified using BN performed significantly better than the original UTAUT. Thus, the model structure

identified by the GS algorithm outperformed the original model and is convincing with regards to its theoretical model structure. As we additionally tested the models versus the GSCA-improved UTAUT, we noted that the linear model based on the GS algorithm could not outperform the GSCA-improved UTAUT, but the structure identified by the HC algorithm did (which provides food for thought in further understanding associations identified by the HC algorithm). The alternative models including the interaction effect and the non-linear relationships outperformed the original UTAUT, but did not outperform the linear version of the GS-improved UTAUT (AM1). A pure metrics-guided assessment would point to the (linear) HC-improved UTAUT (AM2) as the best model. However, combined with the theoretical plausibility assessments we performed, we believe that the GSCA-improved UTAUT performs the best, followed by the GS-improved UTAUT.

Table 9

Model comparisons along predictive accuracy considerations.

Model, constructs	#paths (thereof: sign., *)	$Q^2_{predict}$	RMSE	R^2	BIC	PLS-SEM loss-IA loss	PLS-SEM loss-LM loss
BENCHMARK MODELS							
Original UTAUT (Venkatesh et al., 2012)							
Overall	5 (5)					−0.247**	0.051*
BI	3 (3)	0.341	0.815	0.353	−288.775	−0.299**	0.051*
USE	2 (2)	0.129	0.937	0.250	−189.367	−0.209**	0.050
GSCA-improved UTAUT (Cho et al., 2022a)							
Overall	9 (7)					−0.294**	0.018
BI	4 (4)	0.380	0.790	0.394	−329.766	−0.332**	−0.006
USE	5 (3)	0.156	0.922	0.279	−198.184	−0.265**	0.004
ALTERNATIVE (LINEAR) MODELS							
AM1: GS-algorithm-improved UTAUT							
Overall	6 (6)					−0.273**	0.025
BI	4 (4)	0.380	0.790	0.394	−329.941	−0.332**	0.018
USE	2 (2)	0.138	0.932	0.253	−191.579	−0.229**	0.030
AM2: HC-algorithm-improved UTAUT							
Overall	11 (11)					−0.157**	0.012
BI	4 (4)	0.246	0.872	0.394	−329.508	−0.215**	0.010
USE	3 (3)	0.086	0.960	0.280	−211.418	−0.130**	0.021
ALTERNATIVE MODELS INCL. NON-LINEARITIES AND INTERACTIONS							
AM3: GS-algorithm-improved UTAUT, including interaction							
Overall	7 (6)					−0.273**	0.025
BI	4 (4)	0.380	0.790	0.394	−329.941	−0.332**	0.018
USE	3 (2)	0.138	0.932	0.258	−189.777	−0.229**	0.030
AM4: GS-algorithm-improved UTAUT, including non-linearities							
Overall	9 (7)					−0.266**	0.033
BI	7 (5)	0.377	0.792	0.402	−320.003	−0.329**	0.022
USE	2 (2)	0.131	0.936	0.252	−190.425	−0.219**	0.041

Note: * $p < 0.05$; ** $p < 0.01$.

Hence, in our illustrative example, the selected ML algorithms indicate relevant improvements to the original UTAUT model, especially the new relationships identified via BN. Also, the illustrations demonstrate that identifying non-linearities and interactions using MARS and GAM can hold value for many further research applications – even while not yielding theoretically plausible improvements in our illustration.

5. Discussion, conclusion, and implications

In our study, we demonstrated the combined use of PLS-SEM and selected ML algorithms and offered a step-by-step guideline that integrates past approaches with new ideas to explore relationships and non-linearities, consolidating new capabilities in PLS-SEM. Thereby, we offer avenues for researchers to make relevant contributions to theory and research using the predictive capabilities of PLS-SEM, BN, MARS and GAM. We are confident that combining PLS-SEM with these ML algorithms can ultimately contribute to theorizing in business research by exploiting the two methods' causal-predictive and causal-exploratory capabilities. As demonstrated, the combined approach could contribute first to identifying completely new relationships, improving existing conceptual models, and comparing and making choices between competing models. We note, however, that the methods are tools that assist the researcher in theorizing endeavors by offering ideas inferred from the data, but ultimately, do not construct the theory for the researcher (Provost & Fawcett, 2013; Kuhn & Johnson, 2013).

Researchers often encounter alternative theoretical explanations for a phenomenon, which give rise to different models with different (or additional) antecedents and/or model relationships. PLS-SEM offers an element of exploration, in which the modeling is “[...] used in a more free-form fashion, supporting the purpose of capturing relationships that are perhaps unknown or at least less formally formulated” (Shmueli, 2010, p. 297). This process can provide input to improve existing conceptual models and to develop theoretical hypotheses that we build from the analyzed data. Also, such modeling can represent the inferences of the conducted research (see Jaeger & Halliday, 1998). However, traditional PLS-SEM algorithms do not examine all possible model

configurations unless the researcher actively does so, which can be time-consuming and demanding. The SEM community has introduced the first approaches to enable a prediction-oriented search for models that demonstrate a high predictive accuracy (Cho et al., 2022a). We conclude that BN are particularly advantageous for this purpose due to their exploratory analysis capabilities without the need for predefined conceptual models, while still permitting the integration of some a priori knowledge.

BN are a complementary tool for generating new hypotheses and growing our insight in research situations characterized by weak or multiple theories, complex models with many potentially influencing factors, and interrelated factors and model contingencies. To some extent, BN can free models from one or all the researcher's causal assumptions based on human judgment, past research designs, and popular models. The capability of BN to highlight plausible alternative structures (Bollen & Pearl, 2013; Pearl, 2012a; Pearl, 2012b) and offer insight into new associations (Pearl, 2012a; Pearl, 2012b) is well-documented. Through the detection of both existent and non-existent relationships between the variables most substantiated by the data (Pearl, 1988; Elwert, 2013), BN can detect potential new causal mechanisms, that may lead to the development of new theoretical models or the refinement of existing models. This advantage seems to be exploited to a larger extent in other disciplines. We can, therefore, use BN algorithms as a powerful exploratory technique that provides useful guidance in building PLS-SEM models, mainly when there is little or tenuous theoretical support in constructing the model in business research.

The capabilities of both PLS-SEM and ML can improve existing conceptual models by comparing the accuracy of models involving new relationships between variables, and by mediating, interacting, and integrating other non-linear effects. In PLS-SEM, these relationships can be tested by modeling alternative relationships between multiple variables in a path diagram. PLS-SEM can test non-linearities, interactions, and mediation effects (see Hair et al., 2018; Basco et al., 2022), but the researcher needs to know and manually introduce the higher-order terms and compare models involving these relationships, which can be a complex task. Instead, MARS and GAM could be used more actively to

Table 10

CVPAT test on competing models.

CVPAT compare construct: BI and USE Model 1	Model 2	Model 1 loss	Model 2 loss	Difference	p-value
ALTERNATIVE (LINEAR) MODELS					
Original UTAUT	AM1: GS-algorithm-improved UTAUT	0.849	0.838	0.010	0.000
Original UTAUT	AM2: HC-algorithm-improved UTAUT	0.849	0.830	0.019	0.000
GSCA-improved UTAUT	AM1: GS-algorithm-improved UTAUT	0.833	0.838	-0.006	0.903
GSCA-improved UTAUT	AM2: HC-algorithm-improved UTAUT	0.833	0.830	0.002	0.006
ALTERNATIVE MODELS INCL. NON-LINEARITIES AND INTERACTIONS					
Original UTAUT	AM3: GS-algorithm-improved UTAUT, including interaction	0.849	0.838	0.011	0.001
AM1: GS-algorithm-improved UTAUT	AM3: GS-algorithm-improved UTAUT, including interaction	0.838	0.838	0.000	0.475
GSCA-improved UTAUT	AM3: GS-algorithm-improved UTAUT, including interaction	0.833	0.838	-0.006	0.882
Original UTAUT	AM4: GS-algorithm-improved UTAUT, including non-linearities	0.849	0.839	0.010	0.000
AM1: GS-algorithm-improved UTAUT	AM4: GS-algorithm-improved UTAUT, including non-linearities	0.838	0.839	-0.000	0.555
GSCA-improved UTAUT	AM4: GS-algorithm-improved UTAUT, including non-linearities	0.833	0.839	-0.006	0.895

identify non-linearities and interactions in existing models, and contribute to improving or refining such models, as has been done in most of the past hybrid PLS-SEM and ANN studies in business research. As discussed above, BN are a promising candidate to identify new associations in the form of mediation (Neapolitan & Jiang, 2007; Pearl, 1988). We advocate that MARS and GAM are good approaches to identify and transparently represent non-linearities and interaction effects in existing structural models (Hastie et al., 2009a; James et al., 2021) that have several advantages over and above ANN. Understanding when linear and non-linear responses occur can advance the ability to predict future outcomes in many fields (e.g., behavioral intention or usage) and can provide input for developing new theoretical mechanisms. Understanding these non-linearities better by using the more transparent MARS and GAM has value in the further development of conceptual models.

While previous studies that used ML before applying PLS-SEM reintegrated their findings into structural or conceptual models, studies that used ML after applying PLS-SEM to identify non-linearities did not reintegrate their findings into PLS-SEM. This reintegration can, as we have demonstrated, be useful in profiting from the algorithm's transparency, and to compare models in the comprehensive and well-established PLS-SEM evaluation standards. Analyzing the BIC scores in PLS-SEM and, more importantly, investigating the more recently proposed CVPAT test procedure provide excellent tools to identify models that demonstrate significantly better predictive accuracy than we had before (Liengard et al., 2021; Sharma et al., 2023). Such analyses also assist in finding orientation between various predictive analytics to assess the predictability of an outcome (i.e., the distance between theory and practice). Thus, translating findings from ML back to PLS-SEM enables further relevant evaluation of the models. Also, this enables the usage of PLS-SEM in follow-up studies using further complementary predictive approaches (such as agent-based modeling, Schubring et al., 2016, to name but one option).

In developing the step-by-step guideline, we faced several practical challenges that we would like to mention, which may offer implications for future research. First, there were two approaches to develop construct scores in previous studies combining PLS-SEM and ML: the universal structural modeling approach that seems to be less popular in business research probably due to the complexity of steps involved, and the more pragmatic approach of simply extracting the latent variable scores from PLS-SEM. The challenge with extracting the scores from PLS-SEM is that they rely on the conceptualized structural model relationships. If these are to be validated in a more open exploratory way, this could produce a bias. We, therefore, proposed that we extract scores based on the factor weighting scheme applied to a full model. We invite future research to simulate the implications of this approach in comparison to other approaches used so far. Second, while re-integrating new model structures identified via BN was quite straightforward,

translating non-linearities into PLS-SEM was at first more cumbersome. The ML algorithms pointed to several non-linearities. The finally proposed combination of MARS and GAM including the integration of the deviance test gave us clearer recommendations on non-linearities that are worth being transferred to PLS-SEM. Still the non-linearities detected in ML demonstrated non-significant model improvements after reintegration into PLS-SEM. We found that both the interaction effect identified by the MARS algorithms, as well as the non-linear relationships did not translate into significant path coefficients in PLS-SEM, at least not on a 5 % level. Recognizing MARS's sensitivity in detecting non-linearities and interaction effects, particularly within specific input value ranges, and considering the limited size of our sample, which might have led to potential overfitting, it remains valuable to delve into the conditions under which non-linearities detected by ML models can translate into significant paths in PLS-SEM.

We are confident that our study contributes to a knowledge transfer between researchers using ML algorithms and PLS-SEM and will enable business researchers familiar with PLS-SEM to profit from ML capabilities to advance their conceptual models.

CRedit authorship contribution statement

Nicole F. Richter: Conceptualization, Data curation, Writing - original draft, Writing - review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Resources, Project administration, Software. **Ana Alina Tudoran:** Conceptualization, Data curation, Writing - original draft, Writing - review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Resources, Project administration, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We offer an appendix with code and will provide the file in a repository (see the supplementary appendix for further details).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbusres.2023.114453>.

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